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Research Article

Simple plant–plant interaction patterns predict structural niche and fitness differences in a multitrophic grassland community

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Understanding how species interactions relate to biodiversity requires linking complex interaction networks to coexistence mechanisms. In diverse communities, interactions span trophic levels and include pairwise and higher-order effects, yet their relative importance for explaining population dynamics remains poorly understood. We analyse demographic data from 36 plant species exposed to six grasshopper species in a two-year grassland experiment. We estimate interaction coefficients using lasso regularisation to identify pairwise plant–plant interactions, direct grasshopper effects, and higher-order interactions associated with plant dynamics. We use structural stability theory to quantify structural niche and fitness differences across three- and four-species plant modules, and relate these quantities to metrics describing plant–plant interaction structure. Direct grasshopper effects and higher-order interactions contributed to explaining plant dynamics, but showed limited associations with the intrinsic growth rates and pairwise plant–plant interactions. Variation in structural niche and fitness differences was instead primarily associated with plant–plant interaction structure: stronger intraspecific relative to interspecific interactions were associated with larger structural niche and fitness differences, whereas a more facilitative interaction balance was associated with smaller values of both. Structural niche and fitness differences among plants were largely explained by simple plant–plant interaction patterns despite the complexity of multitrophic interactions.

Keywords: herbivory, higher-order interactions, interaction metrics, lasso regularisation, species coexistence, structural stability

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Introduction

Understanding how species interactions shape the maintenance of biodiversity is a central goal in ecology. A large body of theory shows that coexistence depends on two fundamental mechanisms: niche differences, which determine the range of conditions under which species can coexist, and fitness differences, which determine differences in species' performance within a community (Chesson 2000a, Saavedra et al. 2017, Barabás et al. 2018). In simple systems, these mechanisms can often be inferred from experiments that estimate pairwise species interactions (Kraft et al. 2015). However, natural communities are composed of many interacting species, and how these coexistence mechanisms emerge from complex interaction networks remains poorly understood (Levine et al. 2017, Godoy et al. 2018).

Translating niche and fitness differences to multispecies communities is challenging because species interactions are rarely limited to pairwise interactions (Levine et al. 2017). Instead, they form networks involving multiple species and diverse interaction types (Melián et al. 2009, Mougi and Kondoh 2012, García-Callejas et al. 2018). These include interactions within trophic levels, such as plant–plant interactions, and across trophic levels, such as herbivory. Moreover, the per capita effect of one species on another can be modified by a third species, generating higher-order interactions (HOIs) (Billick and Case 1994, Levine et al. 2017, Kleinhesselink et al. 2022). While theoretical models have increasingly highlighted the potential of these higher-order effects to stabilise complex communities (Bairey et al. 2016, Grilli et al. 2017, Mayfield and Stouffer 2017), empirical research has mostly explored them through the lens of trait-mediated indirect interactions or herbivore-mediated competition in relatively simple modules (Werner and Peacor 2003, Ohgushi 2005, Kaplan and Denno 2007, Kim et al. 2013). More recently, studies have begun to scale up, demonstrating that HOIs fundamentally alter population dynamics in diverse plant communities (Xiao et al. 2020) and providing early experimental evidence that they can shape population dynamics and species persistence within multitrophic networks (Barbosa et al. 2023, Buche et al. 2024). Yet, simultaneously quantifying interaction modifications within trophic levels and across trophic levels, alongside baseline pairwise interactions, presents a considerable experimental and statistical challenge in natural systems. Consequently, the relative importance of pairwise interactions and HOIs in driving overall community dynamics remains poorly understood.

A major obstacle to addressing this challenge is the high dimensionality of ecological interaction networks. In diverse communities, the number of potential interactions, including pairwise effects and HOIs, quickly exceeds what can be reliably estimated from empirical data (May 1972, Weiss-Lehman et al. 2022). As a result, studies often rely on simplified systems or impose strong theoretical assumptions about interaction structure, such as fixing interaction signs or excluding higher-order effects entirely (Bastolla et al. 2009, Rohr et al. 2014). However, these assumptions frequently fail to reflect natural systems, where interaction strengths and signs can vary widely depending on the identities of the interacting species and the ecological context. For example, herbivores are typically assumed to reduce plant performance, yet their negative effects can vary with plant defence strategies (Feeny 1976, Rhoades and Cates 1976), and can be partially offset by compensatory growth responses or nutrient recycling (McNaughton 1983, Agrawal 2000, Garcia and Eubanks 2019). In addition, ecological communities often encompass a mixture of positive and negative interactions. Empirical studies demonstrate that biodiversity is often maximised when both signs coexist within community modules (Godoy et al. 2018, Losapio et al. 2021, Buche et al. 2025). Furthermore, the distribution of these interaction strengths is known to influence system dynamics. For example, system stability is often promoted by a configuration of many weak and few strong interactions (McCann et al. 1998, Berlow 1999, Neutel et al. 2002), or conversely, when interaction strengths are relatively homogeneous across species (Kokkoris et al. 2002). Taken together, these structural constraints highlight the critical need for analytical approaches capable of both estimating interaction strengths empirically and isolating the subset of interactions that meaningfully drives community dynamics.

Even when interaction networks can be estimated, a second challenge remains: understanding how their structure translates into coexistence mechanisms in multispecies systems. Structural stability theory provides a framework to address this by representing these mechanisms geometrically (Saavedra et al. 2017, Song et al. 2018). In this framework, species interactions define a feasibility domain, a region of parameter space corresponding to all combinations of intrinsic growth rates that allow species to coexist. The geometry of this domain captures structural niche differences, while the position of the intrinsic growth-rate vector relative to this domain captures structural fitness differences, which describe asymmetries in species' demographic performance within the interaction structure (see Box 1 for a conceptual primer). Importantly, this geometric interpretation allows structural niche and fitness differences to be quantified in multispecies communities without restricting the sign or type of species interactions (Godoy et al. 2026). However, how structural niche and fitness differences relate to empirically observed interaction structures remains poorly explored.

Here, we analyse demographic data on 36 plant species from a two-year experiment in a diverse grassland where we manipulated the diversity and composition of six grasshopper species. This system provides a unique opportunity to quantify interaction strengths within and across trophic levels under controlled conditions. We use a two-step approach that combines empirical data, statistical inference, and structural stability theory. First, we estimate interaction coefficients using lasso regularisation (Lim and Hastie 2015). This allows us to reduce the dimensionality of the system, identify the subset of interactions that meaningfully contributes to plant population dynamics, and evaluate whether complex interaction types, such as herbivory and HOIs, substantially modify

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Box 1. Structural stability approach

The structural stability approach describes feasibility as the set of intrinsic growth rates compatible with positive abundances for all species, given their interaction structure (Saavedra et al. 2017, Song et al. 2018).
For three species, growth rates can be represented within a simplex (Fig. B1a), where each point corresponds to a particular combination of relative growth rates of species. Only a subset of this space allows feasibility: the feasibility domain (FD) (Fig. B1b), which is fully determined by species interactions. The same geometric interpretation extends to communities with any number of species.
In this framework:

- Structural niche differences (SND) are measured by the size of the FD. Larger SND indicate that feasibility is possible under a wider range of growth-rate combinations, reflecting stronger niche differentiation.
- Structural fitness differences (SFD) describe the position of the intrinsic growth-rate vector (r) relative to the FD. When r lies well within the feasibility domain (i.e. near its centroid), SFD are small; as r approaches the boundary, SFD increase; outside the domain, the community is not feasible.

Whether species can coexist depends on both: structural niche differences determine how large the feasibility domain is, while structural fitness differences determine whether the observed growth rates fall within it.
As an example, Fig. B1c1–c3 shows three hypothetical communities containing the same species, Sp1, Sp2 and Sp3. The first two communities have relatively large feasibility domains, corresponding to larger structural niche differences. However, r is farther from the centroid of the feasibility domain in c1 than in c2, representing larger structural fitness differences. The third community, c3, has the same intrinsic growth rates as c2 but a smaller feasibility domain, corresponding to smaller structural niche differences and a narrower set of compatible intrinsic growth-rate combinations.

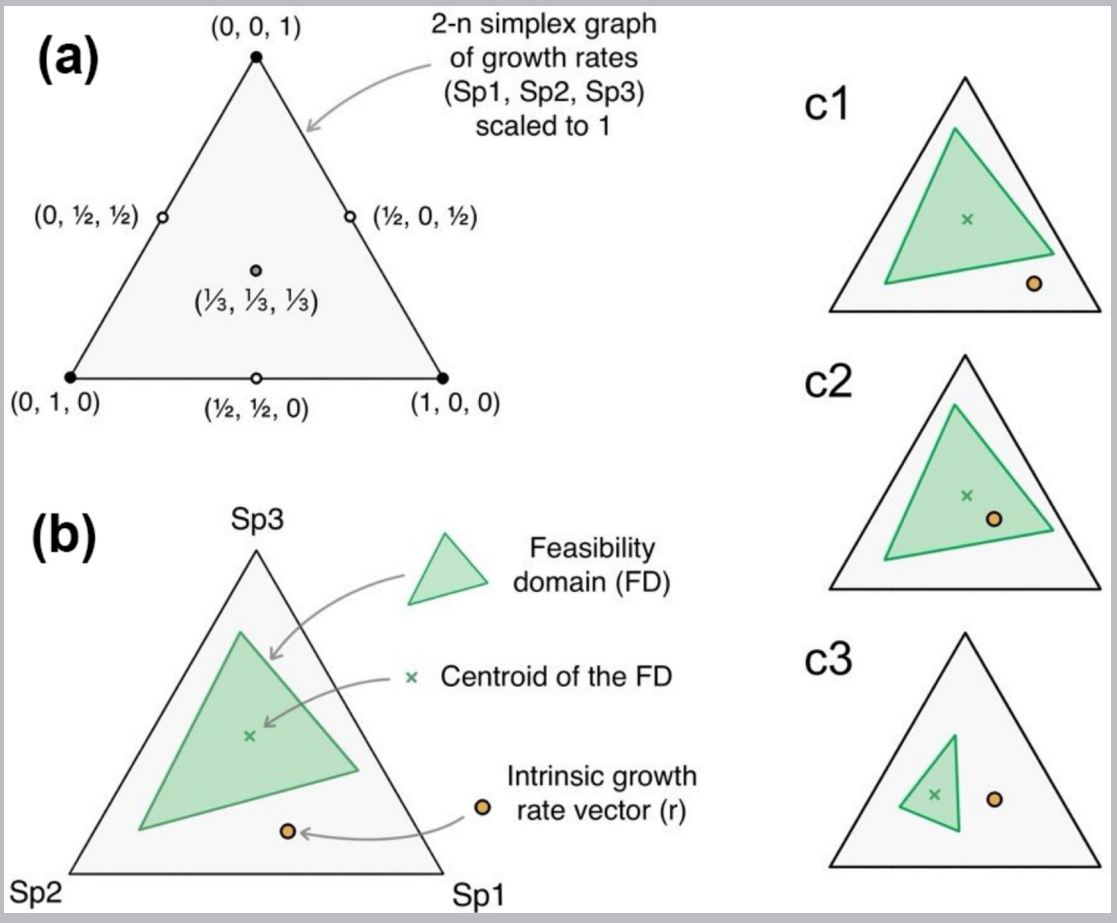


Figure B1. (a) Representation of intrinsic growth rates (r) within a simplex for a three-species community. (b) Feasibility domain (FD; green area, centroid marked by ×) determined by species interactions, and intrinsic growth-rate vector (r; orange point). (c) Three examples illustrating how the position of r relative to the FD determines whether species coexist. The size of the FD reflects structural niche differences, whereas the position of r relative to it reflects structural fitness differences.

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intrinsic growth rates and pairwise plant–plant interactions. Specifically, we expect direct herbivory to be associated with lower plant intrinsic growth rates, and test whether HOIs modify the strength of pairwise plant–plant interactions.

Second, we apply structural stability theory to examine how empirically derived plant–plant interaction networks relate to structural niche and fitness differences in multispecies plant assemblages. Specifically, we calculate structural niche and fitness differences for all three- and four-species plant modules, defined as analytically constructed subcommunities derived from the baseline plant–plant interaction matrix. We use these modules to ask whether variation in structural niche and fitness differences across possible multispecies plant assemblages is associated with four complementary metrics describing the structure of pairwise plant–plant interaction matrices: 1) the balance between intra- and interspecific interactions, 2) the balance between positive and negative interactions, 3) the skewness, and 4) the kurtosis of interaction strengths. We expect stronger intraspecific relative to interspecific interactions and an overall balance between positive and negative interactions to be associated with larger structural niche differences across assemblages (Godoy et al. 2026). We also expect interaction-strength distributions characterised by high kurtosis, that is, many weak and few strong interactions, to be associated with larger structural niche differences, whereas asymmetries in interaction strengths (skewness) should influence structural niche and fitness differences depending on the dominant interaction sign. Thus, rather than evaluating realised coexistence outcomes directly, our analysis asks whether empirically derived plant–plant interaction structures are associated with structural niche and fitness differences as mechanisms expected to regulate multispecies coexistence.

Material and methods

Study system

The experiment was set up for two full years (from June 2012 to May 2014) in a species-rich calcareous grassland (76 plant species in total, 65 recorded in our experiment) located in the LTER 'Zone Atelier Plaine et Val de Sèvre' (46°11′00″N, 0°28′00″W), in central-western France (Deraison et al. 2015a) (Supporting information). Established over 20 years ago on shallow soil, the grassland is managed extensively (no fertilisation, cut once or twice a year), which resulted in a prairie dominated by perennial grasses (average cover of 38.5%), forbs (30.5%), and legumes (7.5%). We focus on the 36 most abundant species, which together represent over 98.5% of plant cover (Supporting information).

Grasshoppers and experimental design

The experiment used a randomised block design (Hurlbert 1984) with five blocks and 14 treatments, totalling 70 cages (Supporting information). Each cage was a 1-m^3 enclosure of transparent insect-proof netting (PE 2230, 920 × 920 μ; DIATEX). These cages created minimal shade and did not significantly alter microclimate (Deraison et al. 2015a). All treatments were randomly assigned within each block. Grasshopper diversity had four levels: 1) control with no grasshoppers; 2) six single-species; 3) six three-species, and 4) one six-species. We selected six species: *Gomphocerippus biguttulus*, *Chorthippus dorsatus*, *Calliptamus italicus*, *Euchorthippus elegantulus*, *Pezotettix giornae* and *Pseudochorthippus parallelus*, because they dominate the area (90% of total abundance) and show wide variation in feeding preferences driven by functional differences in incisor strength (Deraison et al. 2015b). These grasshoppers can remove up to 60% of biomass in summer when adults peak. The three-species treatments were designed to maximise functional diversity of incisor strength relative to plant toughness (Laliberté and Legendre 2010). Grasshopper density was fixed at 24 individuals m^{-2} in all cages, with a constant 1:1 sex ratio, following prior observations (Deraison et al. 2015b). Thus, we manipulated grasshopper diversity and composition, but not plant diversity. Plant cover and composition instead varied naturally across quadrats and time, providing the variation used to estimate plant–plant interactions and plant-mediated interaction modifications. The grasshopper manipulation was not intended to test the effects of grasshopper taxonomic diversity per se, but to generate variation in herbivore identity and feeding preferences to estimate species-specific grasshopper effects on plants, including grasshopper-mediated interaction modifications. Total grasshopper density was kept constant across treatments to isolate identity and composition effects from abundance.

Experimental procedure

The experiment was conducted on the same cages over two consecutive annual transitions (June 2012–May 2013 and June 2013–May 2014) following the same design and procedures (Supporting information). Each annual cycle began in June with a botanical survey to assess the initial plant community in all 70 cages. We visually estimated all species and their cover in nine 10 × 10 cm quadrats evenly spaced within the central region of each cage to avoid edge effects; the quadrat served as our sampling unit rather than the whole cage.

Following the initial survey, we collected 1560 young adults of each grasshopper species from neighbouring grasslands, sexed them, and assigned them to cages. Before introducing the grasshoppers, we cleared aboveground invertebrates from the cages using a vacuum. Throughout the summer period (June–August), we regularly monitored grasshopper density, hand-removed other invertebrates (e.g. spiders), and replaced dead grasshoppers to maintain constant densities. In September of each year, we removed all grasshoppers by hand. The annual cycle concluded the following May with a final botanical survey at the peak of plant growth. While cage frames remained in place year-round, the insect-proof nets were only installed during the summer grasshopper treatments to allow for natural environmental conditions the rest of the year.

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Estimation of interaction coefficients for the full network

Our experimental design, with plants growing without, with one, with three, or with six grasshopper species, allows us to estimate the elements that describe interaction dynamics in our grassland community (Fig. 1). According to a Lotka–Volterra model (Eq. 1), these are: the vector of plant intrinsic growth rates, r (growth in the absence of interactions); the plant–plant interaction matrix, α (intra- and interspecific pairwise interactions); herbivory γ (direct effects of grasshoppers on plants); and two types of HOIs, β (plant-mediated and grasshopper-mediated effects on plant–plant interactions). Positive interaction coefficients indicate facilitative effects, whereas negative coefficients indicate competitive effects. The model is

$$\frac{N_{i,t+1}}{N_{i,t}} = r_i + \sum_{z=1}^{m} \gamma_{iz} N_{z,t} + \sum_{j=1}^{n} \alpha_{ij} N_{j,t} + \sum_{j=1}^{n} \sum_{\substack{k=1 \\ k \neq j}}^{n} \beta_{ijk} N_{j,t} N_{k,t} + \sum_{j=1}^{n} \sum_{z=1}^{m} \beta_{ijz} N_{j,t} N_{z,t}, \quad (1)$$

where $N_{i,t+1} / N_{i,t}$ is the change in cover of plant species i; $N_{j,t}$ and $N_{k,t}$ are the cover of interacting plants j and k; $N_{z,t}$ is the abundance of grasshopper z; r_i is the intrinsic growth rate of i; α_{ij} is the effect of j on i (representing intraspecific interactions when $i = j$ and interspecific interactions when $i \neq j$); β_{ijk} is the HOI mediated by plant k on the effect of j on i; and β_{ijz} is the HOI mediated by grasshopper z on the effect of j on i. We used the Lotka–Volterra model because it is fully compatible with the structural stability framework (AlAdwani and Saavedra 2019). Although Lotka–Volterra models often use abundance or biomass, plant cover change over time is a suitable proxy for modelling interactions (Damgaard and Weiner 2021).

Given the high dimensionality of our approach, we used group lasso regularisation (Tibshirani 1996, Bakin 1999, Yuan and Lin 2006) via *glinternet.cv* in the 'glinternet' package (Lim and Hastie 2015) to reduce the number of selected variables. Briefly, lasso adds a penalty to the objective function of the model, producing sparsity by shrinking weakly supported coefficients to zero. We use selected or retained coefficients to refer to terms that remain non-zero after regularisation, whereas non-selected candidate terms are shrunk to zero. Sparsity is controlled by the tuning parameter λ: when $\lambda = 0$, no sparsity is added and all coefficients are retained; as λ increases, more coefficients are shrunk to zero. Group lasso is hierarchical, meaning that an HOI can only be retained if its corresponding lower-order terms are also retained. For example, a plant-mediated HOI β_{ijk}, in which plant k modifies the effect of plant j on plant i, can only be retained if the lower-order effects of plant j and plant k on plant i are also retained. Similarly, a grasshopper-mediated HOI β_{ijz} can only be retained if the plant–plant effect α_{ij} and the direct grasshopper effect γ_{iz} are also retained. Thus, group lasso improves interpretability by retaining supported variables and shrinking unsupported coefficients to zero. This is critical in our multispecies community because a non-penalised approach would require estimating up to 55 980 coefficients across growth rates, direct interactions, and HOIs, which is infeasible for most empirical datasets.

For each of the 36 plant species, we fitted a statistical model matching the Lotka–Volterra notation (Eq. 1) with lasso regularisation: the response was the focal species' plant cover, and predictors were neighbour plant cover per quadrat and the abundance of each grasshopper species in the cage. We used plant cover data from June 2012, May 2013, and May 2014 to fit the model.

To select the tuning parameter λ, we used cross-validation with *glinternet.cv*, fitting 50 models with decreasing λ. Larger λ yields greater shrinkage towards zero. We selected the λ that minimised cross-validation error and retained the intraspecific interaction for each focal species, which is key for coexistence analyses (Saavedra et al. 2017, Song et al. 2018), and for evaluating some of the metrics proposed in our analyses such as the balance between intra- and interspecific interactions. Species-specific λ values (range = 0.0066–0.2089) are listed in the Supporting information. We used species-specific λ rather than a shared one, which would be excessively high for most species and retain non-essential coefficients, defeating the purpose of lasso. This approach allowed us to estimate the intrinsic growth vector r and the interaction matrices α, γ, and β (both β_α and β_γ), which describe our interaction network. This approach uses the variation generated by the grasshopper treatments to estimate intrinsic growth rates and pairwise plant–plant interactions under realistic multitrophic conditions, which form the basis of the subsequent coexistence analysis.

To assess data sufficiency for this high-dimensional model, we plotted per capita growth of focal species against neighbour density (Hart et al. 2019). Data were insufficient to support coefficient estimation for about one-third of species combinations. A key strength of our approach is that regularisation naturally handles these cases by shrinking unsupported coefficients toward zero. For the remaining interactions, we found no indication of data limitations (Supporting information).

Uncertainty and the importance of interaction types

Lasso regularisation does not directly provide coefficient standard errors (Tibshirani 1996), which limits uncertainty propagation. To overcome this, we quantified uncertainty via bootstrapping (Meyer et al. 1986). For each species, we resampled our empirical data with replacement to obtain a new dataset of the same size, fitted the lasso, and repeated the process 1000 times. As in the empirical fit, each bootstrap selected the λ that minimised cross-validation error and estimated intraspecific interactions. We removed 34 bootstraps that failed to estimate intraspecific interactions within the predetermined λ range, leaving 966 bootstrapped coefficient sets. Using the empirical and

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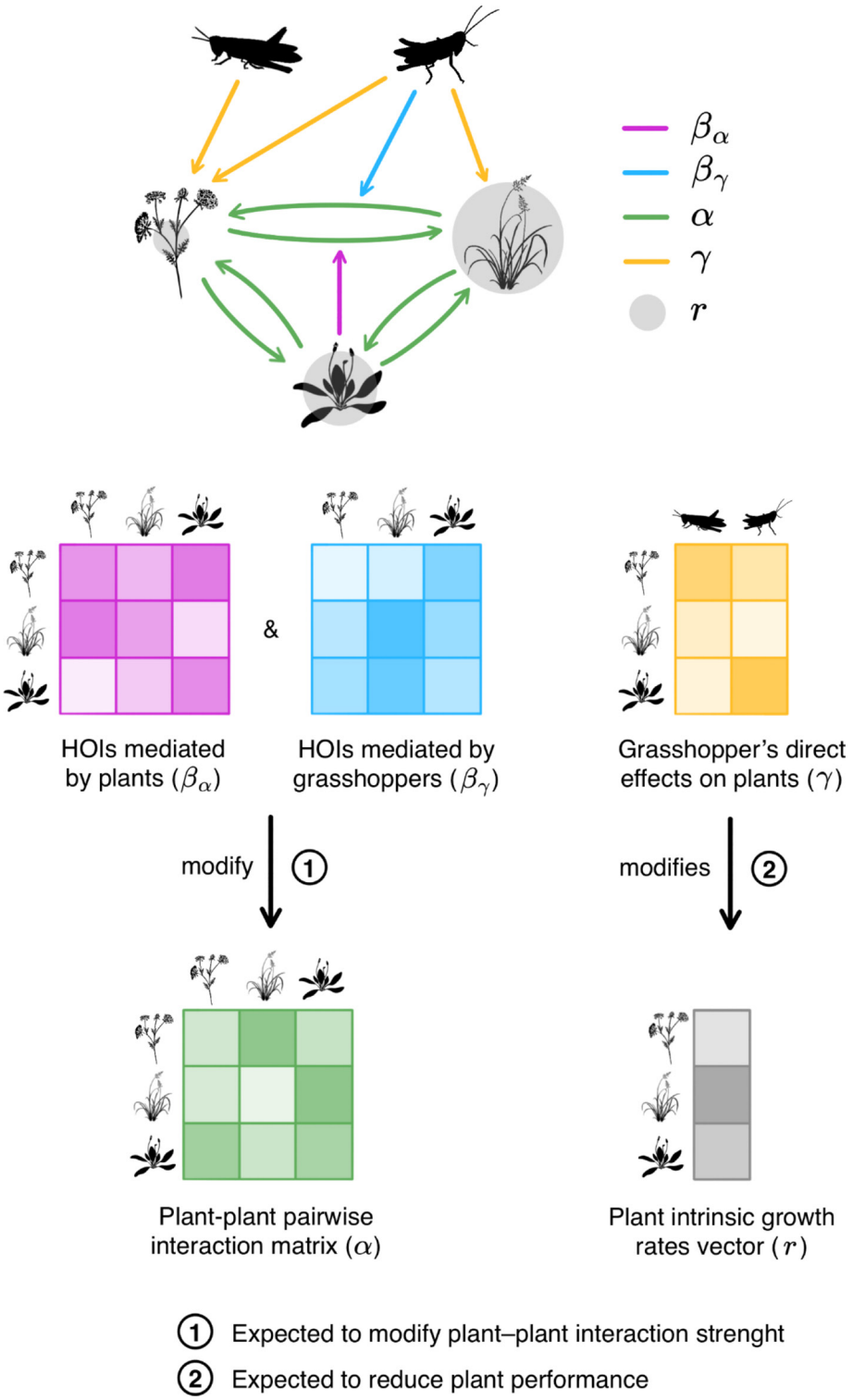


Figure 1. Conceptual framework linking higher-order interactions (HOIs) and herbivory to plant interaction structure. Grasshoppers influence plant communities through two pathways. First, HOIs mediated by plants (β_α) and grasshoppers (β_γ) modify pairwise plant–plant interactions (α). Second, grasshoppers exert direct effects on plants (γ), which alter plant intrinsic growth rates (r). These processes determine the structure of the interaction matrix (α) and the growth-rate vector (r). The network illustrates the different types of interactions (colours), with colours corresponding to the matrices below, where lighter shades indicate weaker interactions and darker shades stronger interactions. These interaction coefficients contribute to the per capita growth rate of a focal plant species in the Lotka–Volterra model (Eq. 1). HOIs are expected to modify plant–plant interaction strength (1), whereas herbivory is expected to reduce plant performance (2). Icons by Rheann Earnest.

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bootstrapped coefficients, we calculated 99% confidence intervals to test whether direct herbivory by grasshoppers modified plant intrinsic growth rates and whether HOIs changed the plant–plant interaction matrix (Fig. 1). We used 99% confidence intervals as a conservative threshold given the high dimensionality of the model and the large number of estimated coefficients.

Structural mechanisms of coexistence

Using the estimated plant–plant (α) interaction coefficients and intrinsic growth rates (r), we calculated structural niche and fitness differences according to the structural stability approach (Box 1) for all possible three- and four-species plant modules. These modules correspond to analytically defined submatrices of the empirical plant–plant interaction matrix, derived from all unique combinations of three and four plant species in the full empirical matrix. We used three- and four-species modules because they provide the smallest tractable multispecies assemblages for evaluating structural niche and fitness differences while remaining biologically interpretable across the larger number of possible species combinations. Three-species modules represent the first level beyond pairwise comparisons, where indirect effects generated by the pairwise interaction matrix can emerge (Ranjan et al. 2024). Four-species modules extend this by allowing a wider range of interaction configurations and more complex network structures. Together, these modules allowed us to test how variation in empirically inferred plant–plant interaction structure is associated with structural niche and fitness differences across possible multispecies plant assemblages.

Metrics describing the configuration of interactions within modules

To link structural niche and fitness differences to the configuration of plant–plant interactions, we characterised each module using four complementary metrics calculated from the baseline plant–plant interaction matrix: intra-interspecific balance (IIB), positive–negative balance (PNB), skewness, and kurtosis of interaction strengths (Fig. 2). These metrics describe whether plant modules differ in the relative strength of intra- and interspecific interactions, the balance between positive and negative interactions, and the shape of the interaction-strength distribution.

IIB quantifies the difference between average intraspecific and interspecific interaction strengths within each module. It was calculated as the mean of the diagonal of the plant–plant interaction matrix, representing average intraspecific effects, minus the mean of the off-diagonal coefficients, representing average interspecific effects across all pairwise plant–plant interactions in the module. Positive values therefore indicate that intraspecific coefficients are larger on average than interspecific coefficients, whereas negative values indicate that interspecific coefficients are larger on average than intraspecific coefficients. We used means, rather than sums, so that IIB described average interaction strength within a module rather than scaling with the number of coefficients. PNB captures the net sign of pairwise plant–plant interactions within each module. It was calculated as the sum of all interaction coefficients, with more positive values indicating modules dominated by facilitative rather than competitive interactions. Skewness measures the asymmetry of the interaction-strength distribution, indicating whether the strongest interspecific interactions tend to be positive or negative. Kurtosis measures the tail weight of the distribution, distinguishing modules with many weak and few strong interactions from modules with more evenly distributed interaction strengths.

We then tested whether variation in structural niche and fitness differences was associated with these four module-level interaction-structure metrics. We fitted generalised linear models (GLMs) with a Tweedie error distribution and a *log-link* function via the 'glmmTMB' package (Brooks et al. 2017), which accommodates right-skewed data that include zeros. Random effects for block or year were not included, as the modules were extracted from a single global interaction matrix. To allow for direct comparison of standardised associations, all predictor variables (IIB, PNB, skewness, and kurtosis) were scaled to zero mean and unit variance prior to analysis.

Our primary analysis evaluated the independent associations with structural niche and fitness differences of all four metrics simultaneously. For both three- and four-species modules, we fitted an additive GLM modelling structural niche differences or structural fitness differences as a function of the scaled metrics (i.e. $\log(E[y]) = b_1\,\text{IIB} + b_2\,\text{PNB} + b_3\,\text{skewness} + b_4\,\text{kurtosis} + b_0$). We extracted the standardised coefficients and 99% confidence intervals to determine the direction and magnitude of each metric's association with structural niche and fitness differences. Finally, we also fitted separate GLMs to explore the associations of each metric individually and in all pairwise combinations; results from these analyses are provided in the Supporting information.

All analyses were performed in R ver. 4.4.3 (www.r-project.org). A full list of the packages used, generated with grateful ver. 0.3.0 (Rodriguez–Sanchez and Jackson 2025), can be found in the Supporting information.

Results

Lasso regularisation identified a sparse interaction structure linking plant–plant and plant–grasshopper interactions to changes in plant cover. The model retained non-zero coefficients for all interaction types, including 49% of pairwise plant–plant interactions (α), 42% of direct grasshopper effects on plants (γ), and 46% of HOIs (β) when pooled across all HOI matrices (Fig. 3). However, individual HOI matrices were much sparser, retaining only 2.5% ± 0.9% of possible coefficients, and the full model retained 2089 non-zero coefficients out of 55 980 possible terms. All interaction types included both positive and negative coefficients, but HOIs were more concentrated around zero (β range = −0.08 to 0.19) than pairwise plant–plant interactions (α range = −0.51 to 0.60) or direct grasshopper effects

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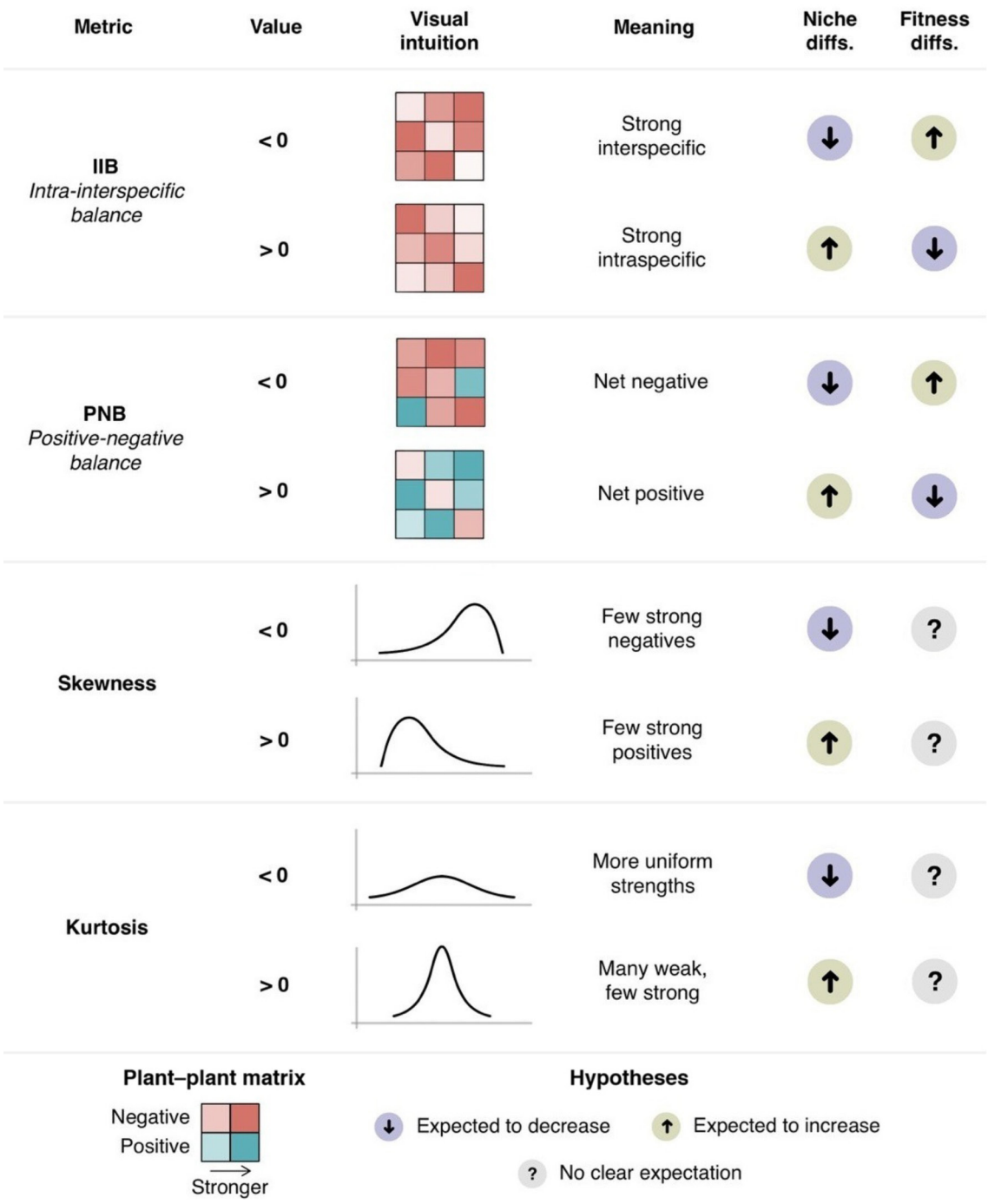


Figure 2. Summary of plant–plant interaction metrics and their directional expectations on structural niche and fitness differences. Four metrics describing the structure of plant–plant interaction matrices are illustrated: intra-interspecific balance (IIB), positive–negative balance (PNB), skewness, and kurtosis. For each metric, values above and below zero correspond to different configurations of interaction strengths, shown schematically ('visual intuition') and interpreted in terms of their ecological meaning. Symbols indicate the expected direction of change in structural niche differences (niche diffs.) and structural fitness differences (fitness diffs.): arrows denote increases or decreases, whereas question marks indicate no clear expectation. Colours denote negative (red) and positive (blue) interactions, with darker shades indicating stronger effects.

(γ range = −0.65 to 0.44), indicating that individual HOI terms were generally weaker.

We initially hypothesised that direct herbivory would strongly influence year-to-year plant intrinsic growth rates, and that HOIs would substantially alter how plants interact with each other. While we observed a positive relationship between the magnitude of plant intrinsic growth rates and herbivory strength (Fig. 4a), this association was not enough

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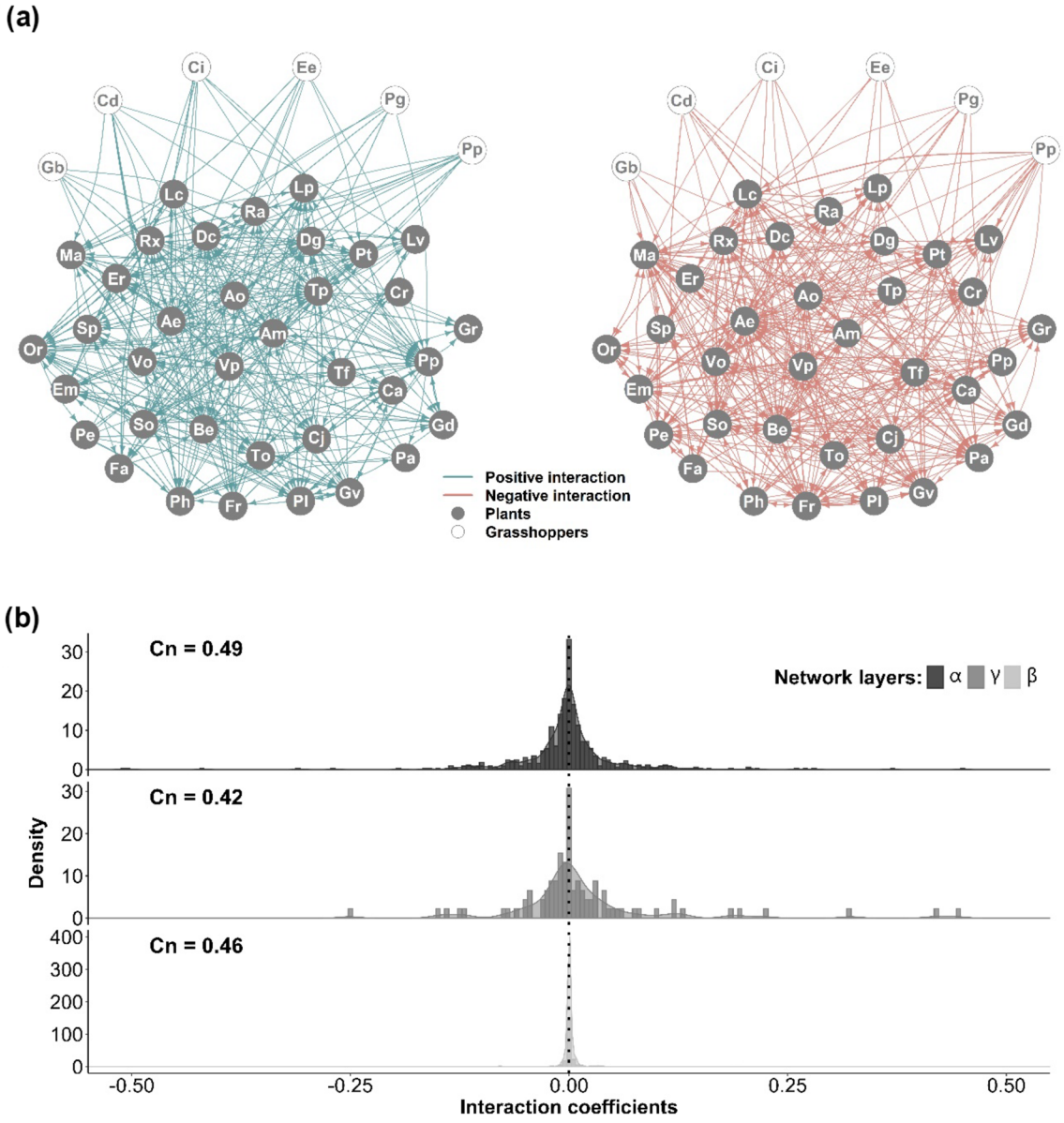


Figure 3. Empirical plant–grasshopper interaction networks and coefficient distributions. (a) Directed networks illustrating positive (facilitative; left) and negative (competitive; right) interactions among the 36 plant species (grey nodes) and 6 grasshopper species (white nodes) evaluated in this study. Edge width is proportional to the absolute magnitude of the interaction coefficient. (b) Density distributions of the non-zero interaction coefficients separated by network layer: pairwise plant–plant interactions (α, top), direct grasshopper effects on plants (γ, middle), and higher-order interactions (HOIs, β, bottom). The β coefficients represent the sum of all plant-mediated and grasshopper-mediated HOIs. The vertical dotted line denotes zero. The calculated connectance (*Cn*), defined as the proportion of realised non-zero interactions relative to all possible interactions within that layer, is provided for each distribution.

to meaningfully modify the plant intrinsic growth rates overall (Fig. 4b). We observed a negative correlation between pairwise plant–plant interactions and their corresponding HOIs (Fig. 4c). Specifically, competitive plant–plant interactions (negative α) were systematically associated with facilitative HOIs, which mathematically buffer or dampen the strength of the baseline pairwise competition (Fig. 4d); however, the confidence interval for the relationship between pairwise plant–plant interactions and HOIs was notably wide ($R^2 = 0.983$ [0.388, 0.998]). These findings held true for the entire community matrix, independent of module size. Consequently, the baseline pairwise plant community remained the dominant interaction layer in our system.

Our analysis revealed that the structure of interactions within modules, characterised by intra-interspecific balance (IIB), positive–negative balance (PNB), skewness, and kurtosis, was strongly associated with both structural niche differences and structural fitness differences. Evaluating all

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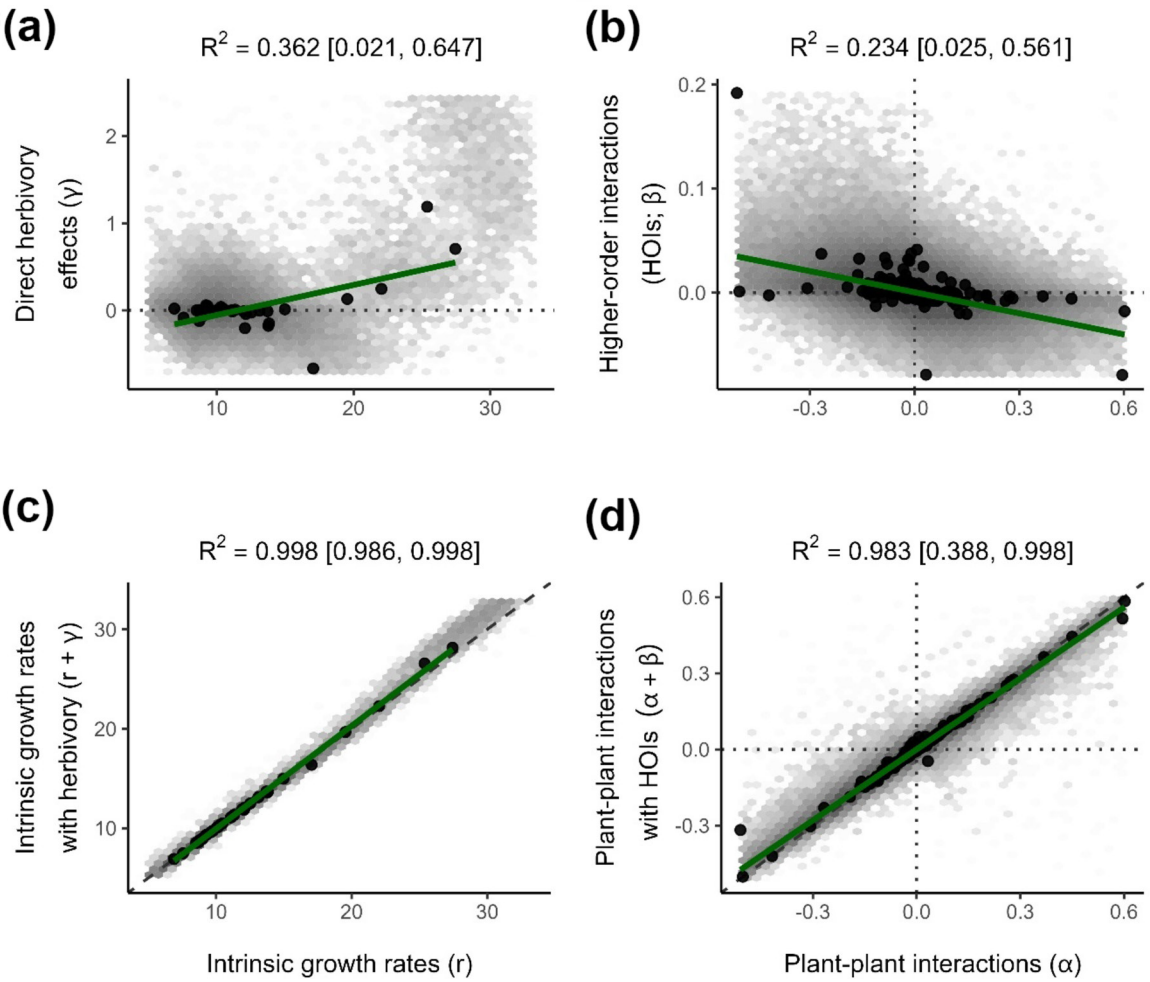


Figure 4. Effects of herbivory and higher-order interactions (HOIs) on plant performance and pairwise interactions. (a) Direct herbivory effects (γ) plotted against baseline plant intrinsic growth rates (*r*). (b) Net intrinsic growth rates incorporating herbivory ($r+\gamma$) compared to baseline intrinsic growth rates. (c) HOI coefficients (β) plotted against baseline pairwise plant–plant interaction coefficients (α). (d) Net plant–plant interaction coefficients incorporating HOIs ($\alpha+\beta$) compared to baseline pairwise interactions. In panels (b) and (d), the dashed grey lines represent a 1:1 relationship (e.g. where the net value equals the baseline value). Across all panels, hexagonal bins (grey shading) represent the density of bootstrapped estimates (darker shades indicate higher density). Black points and solid green lines show empirical point estimates and their corresponding linear regressions. Plot titles display the empirical R^2 alongside its bootstrap-derived 99% confidence interval. Species-specific growth rate estimates are provided in the Supporting information.

structural metrics through an additive GLM (Fig. 5), IIB and PNB emerged as the most robust and consistent correlates of structural niche and fitness differences. Higher IIB (representing stronger intraspecific interactions relative to interspecific interactions) was consistently associated with larger structural niche and fitness differences, whereas higher PNB (indicating a greater proportion of positive or facilitative interactions) was associated with smaller values of both coexistence metrics.

The distribution of interaction strengths, captured by skewness and kurtosis, showed more nuanced associations with structural niche and fitness differences that often depended on module size. Higher skewness was positively associated with structural niche differences in both three- and four-species modules, whereas its association with structural fitness differences was negative in three-species modules but positive in four-species modules. The absolute magnitude of most modelled associations was larger in four-species modules than in three-species modules (Fig. 5). These patterns were consistent when structural metrics were evaluated individually and in pairwise combinations (Supporting information). Across these supplementary analyses, IIB and PNB remained the strongest correlates of structural niche and fitness differences, with IIB showing consistently positive associations and PNB consistently negative associations.

Discussion

Ecologists have long sought to understand the role of species interactions in maintaining biodiversity, yet estimating these interactions in diverse, multitrophic systems remains

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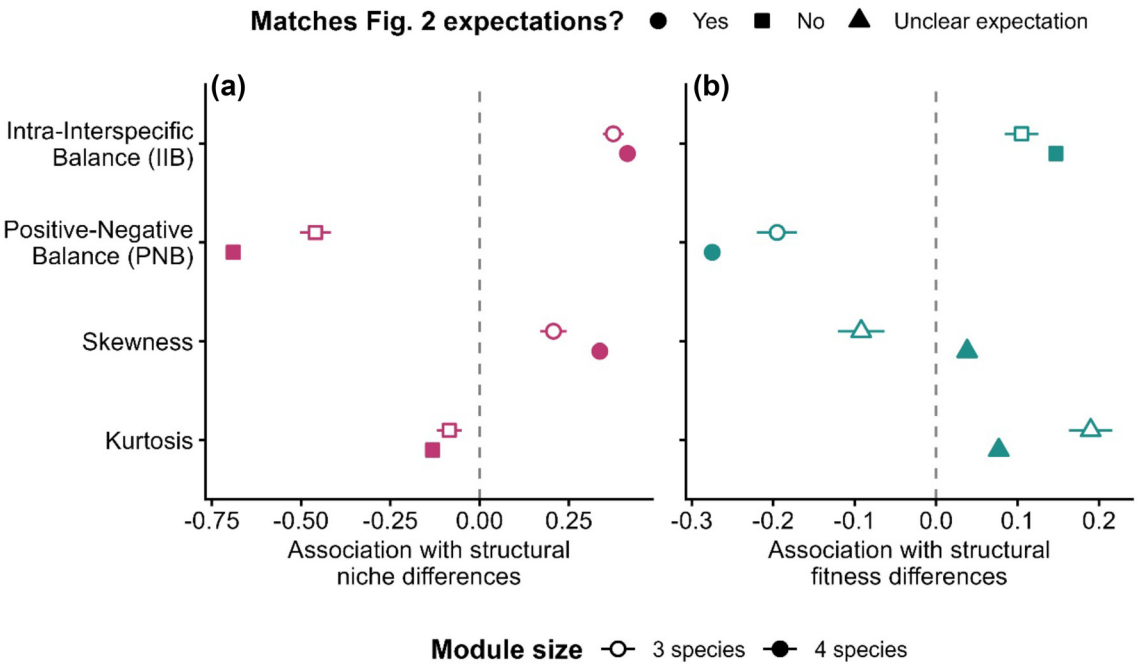


Figure 5. Additive associations between interaction structure and structural niche and fitness differences. Standardised coefficients from generalised linear models with a Tweedie distribution and log-link evaluating the simultaneous associations of four interaction metrics (intra-interspecific balance, IIB; positive–negative balance, PNB; skewness; and kurtosis) with (a) structural niche differences (SND) and (b) structural fitness differences (SFD). Points represent estimated partial associations on the log-link scale, corresponding to the change in the log expected response per standard deviation increase in each predictor while controlling for all other metrics. Horizontal lines indicate 99% confidence intervals. Models were fitted independently for three-species (white fill) and four-species (solid fill) interaction modules. The vertical dashed grey line at zero indicates no association on the log-link scale. All predictor variables were scaled prior to analysis to compare relative associations among metrics within models. Baseline values (model intercepts on the log-link scale) for average 3-species modules were 0.01 [−0.01, 0.04] for SND and 3.82 [3.80, 3.84] for SFD; for four-species modules, intercepts were 0.01 [−0.01, 0.02] for SND and 3.97 [3.80, 3.97] for SFD. Shapes indicate whether the direction of the associations matches the expectations posed in Fig. 2. Results from models evaluating pairwise associations between metrics and structural niche and fitness differences are provided in the Supporting information.

challenging because the number of potential interactions rapidly exceeds what can be reliably inferred from empirical data (Levine et al. 2017, Godoy et al. 2018). Here, we used regularisation to shrink weakly supported interactions towards zero and identify the interactions associated with temporal variation in plant cover, following similar applications in annual plant communities (Weiss-Lehman et al. 2022) and microbial systems (Coyte et al. 2021). Interaction coefficients were retained across all interaction types, including pairwise plant–plant interactions, direct grasshopper effects, and HOIs, indicating that each interaction type contributed to explaining plant dynamics. However, most potential interactions were not retained, especially within individual HOI matrices. This sparsity should be interpreted in relation to the model-selection role of lasso regularisation. Whereas coefficients retained by the model identify supported interaction terms, coefficients shrunk to zero correspond to candidate interaction terms that were not retained by the regularised model, rather than interactions that can be assumed to be biologically absent. Within this detectable interaction structure, plant dynamics were largely captured by plant intrinsic growth rates and pairwise plant–plant interactions, whereas direct grasshopper effects and HOIs showed more limited associations with these core plant demographic and interaction components. This predominance of pairwise plant–plant interactions may reflect their broader variation in strength and sign, as previously reported in empirical and theoretical studies (Rossberg et al. 2011, Gellner and McCann 2016, Barabás 2021).

Direct grasshopper effects require a separate interpretation because they contributed to explaining plant dynamics but showed limited associations with plant intrinsic growth rates. This is notable given that grasshoppers can remove up to 60% of aboveground plant biomass in this system (Deraison et al. 2015a). At first glance, such substantial biomass loss would be expected to translate into strong effects on population growth (Allan and Crawley 2011, Mortensen et al. 2018). However, this pattern can be explained by a phenological mismatch and the temporal scale at which these demographic processes operate. Grasshopper abundance peaked during the summer when they were introduced into the experiment, at which point they primarily consumed mature or senescing tissues after the most rapid spring growth of these perennial plant species had already occurred (Deraison et al. 2015a,

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2015b). Because intrinsic growth rates are estimated over full annual transitions, they integrate demographic performance across the entire interannual cycle rather than short-term seasonal biomass loss. In addition, the dominant perennial grasses in this system are well adapted to recurrent herbivory, with strong tolerance to defoliation. This buffering capacity is consistent with compensatory responses, and in some cases overcompensation (McNaughton 1983, Agrawal 2000, Garcia and Eubanks 2019), in which plants offset tissue loss through regrowth or resource reallocation from belowground reserves. Importantly, such responses mitigate the negative effects of herbivory on growth but do not necessarily imply positive effects on fitness. As a result, herbivory can substantially reduce biomass within a season while having limited influence on interannual population growth rates.

HOIs, when detected, were consistent with a partial buffering of pairwise plant–plant interactions, shifting both competitive and facilitative effects towards zero. This suggests that the ecological effect of one plant species on another was not fixed, but depended on the surrounding plant and herbivore context. For competitive pairwise interactions, positive HOIs indicate that additional neighbours or herbivores can partly relax the negative effect of a competitor, whereas for facilitative pairwise interactions, negative HOIs indicate that the positive effect of a neighbour can be weakened in more complex biotic contexts. Such context dependence may arise through several non-mutually exclusive processes. Neighbouring plants can alter focal apparency, herbivore pressure, or trait expression, thereby changing the realised strength of plant–plant interactions (Werner and Peacor 2003, Ohgushi 2005, Kim et al. 2013, Underwood et al. 2014). Similarly, grasshoppers may modify plant–plant interactions through species-specific feeding preferences, while neighbouring plant identity and composition can generate associational resistance or susceptibility (Kim et al. 2013, Underwood et al. 2014, Deraison et al. 2015b, Sotomayor and Lortie 2015). This interpretation differs between plant- and grasshopper-mediated terms because grasshopper composition was experimentally manipulated, whereas plant cover and composition varied naturally. Grasshopper-mediated HOIs are therefore linked to experimentally generated variation in herbivore composition, whereas plant-mediated HOIs describe interaction modifications associated with natural variation in plant cover and composition. Importantly, our phenomenological Lotka–Volterra framework captures these net interaction modifications without resolving their underlying mechanisms. More recently, studies have begun to quantify HOIs at the community level in species-rich systems (Xiao et al. 2020, Barbosa et al. 2023, Shen et al. 2023, Buche et al. 2024, 2025), showing that interaction modifications can be widespread and context-dependent. In our case, however, higher-order terms were weak and sparse relative to pairwise plant–plant interactions. This pattern may partly reflect the difficulty of estimating higher-order terms in species-rich communities, where the number of potential interaction modifications increases combinatorially with community size and reduces the statistical power to detect individual coefficients (Levine et al. 2017, Mayfield and Stouffer 2017, Kleinhesselink et al. 2022). Nevertheless, the limited magnitude and frequency of detected higher-order terms suggest that they played a comparatively minor role in explaining community dynamics relative to pairwise plant–plant interactions.

We next examined how the structure of plant–plant interactions relates to structural niche and fitness differences in multispecies communities. For this analysis, we focused on analytically defined three- and four-species modules derived from the baseline plant–plant interaction matrix, representing the smallest tractable multispecies sub-communities. Across these modules, structural niche and fitness differences varied systematically with key features of the interaction matrix. In particular, the balance between intra- and interspecific interactions and the balance between positive and negative interactions showed consistent associations with structural niche and fitness differences, whereas properties of the interaction strength distribution, such as skewness and kurtosis, showed more variable associations. These results indicate that relatively simple aspects of plant–plant interaction structure are associated with the range of structural niche and fitness differences observed in multispecies assemblages.

The structure of interactions was strongly associated with structural niche and fitness differences, although not always in the way initially anticipated. When interactions within species were stronger than those between species, structural niche differences were consistently larger, indicating that the relative balance of within- versus between-species interactions is closely linked to stabilising potential, as expected from coexistence theory (Chesson 2000b, Chesson and Kuang 2008, Barabás et al. 2016). In our analyses, this association emerged across multispecies modules containing both facilitative and competitive effects (Mougi and Kondoh 2012). At the same time, stronger within-species interactions were associated with larger structural fitness differences. This association can also be understood by considering that our multispecies modules contain a mix of competitive and facilitative interactions. When strong within-species interactions occur in the presence of facilitation, the range of conditions allowing species to persist can expand beyond the boundaries defined by competition alone (Mougi and Kondoh 2012), potentially increasing the distance between the feasibility domain's centroid and the vector of intrinsic growth rates rather than reducing it (Godoy et al. 2026).

A similarly informative pattern emerged for the balance between facilitative and competitive effects. Communities dominated by facilitative interactions showed reduced structural fitness differences, consistent with the idea that facilitative interactions can homogenise species performance. However, these same communities also exhibited lower structural niche differences, suggesting that niche differentiation may be maximised not under net facilitation, but in systems where facilitative and competitive interactions are more balanced and partially offset one another (Gross et al. 2015, Losapio et al. 2021). In empirical terms, modules dominated by facilitative interactions may reduce differences in species'

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demographic performance, thereby lowering structural fitness differences, but may also make the interaction environment more uniformly positive across species. By contrast, modules containing a balance of facilitative and competitive interactions may create more heterogeneous interaction environments, allowing species to differ more strongly in how they respond to and affect one another. For both interaction-balance metrics, IIB and PNB, the strength of the observed associations increased from three- to four-species modules, suggesting that the relationship between interaction structure and structural niche and fitness differences becomes more pronounced as community complexity increases.

The distribution of interaction strengths also covaried with structural niche and fitness differences, although in a more context-dependent way. Positive skewness, corresponding to a few strong facilitative interactions embedded within many weaker competitive ones, was associated with large structural niche differences, consistent with previous work showing that facilitation can promote diversity under specific configurations (Gross 2008, Johnson 2021, Losapio et al. 2021). However, the correlation between positive skewness and structural fitness differences depended on community size, being negative in three-species modules but positive in four-species modules, indicating that asymmetries in interaction strength do not translate into consistent patterns of species performance. Kurtosis, reflecting the balance between many weak and few strong interactions, was also associated with variation in structural niche and fitness differences. Contrary to our expectations, higher kurtosis was associated with slightly lower structural niche differences, suggesting that this feature alone does not necessarily enhance stabilising processes. For structural fitness differences, kurtosis was the only interaction-structure metric whose association was weaker in four-species than in three-species modules, highlighting that the relationship between interaction-strength distributions and coexistence mechanisms can vary with community size. Taken together, these results show that, beyond the overall balance of facilitative and competitive interactions, the way interaction strengths are distributed across species adds an additional layer of structure associated with structural niche and fitness differences, particularly as communities become more diverse.

The key findings of our study are based on a regularisation approach used to estimate interactions in a high-dimensional system. We calibrated this approach with data that capture natural variation and temporal dynamics in plant cover across two annual transitions, while experimentally manipulating the herbivore community. This combination of experimental control and statistical regularisation was essential to make inference feasible. However, it also presents a challenge in fully capturing system dynamics. Extending the temporal scope of the study by following communities over more annual transitions could reveal additional variation in interaction strengths and signs, including variation associated with environmental fluctuations such as precipitation or nutrient inputs (Granjel et al. 2023, Hallett et al. 2023). Our results should therefore be interpreted as identifying the subset of interactions that can be robustly inferred from the available data and model structure, rather than providing an exhaustive characterisation of all possible interactions in the system.

Our modelling framework assumes a specific functional form to describe species interactions, which is a necessary simplification to make the problem tractable. Although our main conclusions regarding structural niche and fitness differences are based on plant intrinsic growth rates and pairwise plant–plant interactions, the demographic models used to estimate plant dynamics also included HOI terms. These higher-order terms can, in principle, lead to differences between pairwise and higher-order representations of community dynamics (Buche et al. 2024). However, the structural stability approach used here calculates structural niche and fitness differences from the intrinsic growth-rate vector and the pairwise plant–plant interaction matrix. For this reason, HOI terms were included when estimating plant dynamics, but not when calculating structural niche and fitness differences. Exploring alternative functional forms and comparing their performance would be a valuable next step (Terry and Armitage 2024), as shown in lower-dimensional systems (Hart et al. 2019), although scaling such approaches to species-rich communities remains challenging given functional trait heterogeneity and uneven data availability.

In addition, our approach focuses on plant species coexistence mechanisms without explicitly considering the joint structural niche and fitness differences involving grasshopper species. Because grasshopper densities were maintained constant throughout the experiment, we did not assess how plant dynamics in turn affect herbivore populations (Godoy et al. 2018). While obtaining such information under natural conditions is challenging, experimental approximations in large cages may help address this limitation (Bartomeus et al. 2021).

While our study quantifies a complex network of species interactions in a multitrophic community, it leads to a relatively simple conclusion. The mechanisms underlying multispecies coexistence in our system were largely captured by intrinsic growth rates and pairwise plant–plant interactions, which form the basis of our structural analysis. Although herbivory and HOIs contributed to explaining plant dynamics, they showed limited associations with plant intrinsic growth rates and pairwise plant–plant interaction strengths and were not directly included in the calculation of structural niche and fitness differences. Instead, variation in structural niche and fitness differences was primarily associated with how interaction strengths were organised within the plant community. In particular, the relative balance between within- and between-species interactions, and between facilitative and competitive interactions, emerged as central correlates across multispecies modules. This suggests that, despite the high dimensionality of ecological networks, their relationship with coexistence mechanisms may be captured by a limited set of structural properties. More broadly, our results point to a potential asymmetry across trophic levels, in which plant interactions define the structure of coexistence mechanisms while herbivores operate within this framework through species-specific

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effects. Whether such differences in the relative importance of interaction types are general across ecosystems remains an open question. By combining experimental data, regularisation techniques, and structural approaches, our study provides a tractable framework for linking complex interaction networks to coexistence mechanisms, and offers a pathway for applying structural coexistence approaches to empirically grounded, species-rich systems.

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Author contributions

Rodrigo R. Granjel: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Funding acquisition (supporting); Methodology (equal); Project administration (equal); Software (lead); Supervision (equal); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review and editing (lead). **Nicolas Gross**: Conceptualization (supporting); Data curation (supporting); Funding acquisition (lead); Investigation (equal); Project administration (equal); Resources (equal); Supervision (supporting); Writing – review and editing (supporting). **Isabelle Badenhausser**: Investigation (equal); Resources (equal); Writing – review and editing (supporting). **Juan Arroyo**: Supervision (supporting); Writing – review and editing (supporting). **Oscar Godoy**: Conceptualization (equal); Formal analysis (supporting); Funding acquisition (supporting); Methodology (equal); Project administration (equal); Software (supporting); Supervision (lead); Writing – original draft (supporting); Writing – review and editing (supporting).

Data availability statement

Data are available from the Dryad Digital Repository: https://doi.org/10.5061/dryad.2rbnzs866 (Granjel et al. 2026), and code are available from Zenodo: https://doi.org/10.5281/zenodo.23054165.

Supporting information

The Supporting information associated with this article is available with the online version.

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